Q1. **Data Understanding** Identify all data quality issues present in the dataset that can cause problems during data loading.

Ans : Issues are :

a) Inconsistent Formats: The Order_Date field uses multiple formats
b) Mixed Data Types: The Sales_Amount column contains numeric values (4500), text-based numbers (Three Thousand), and Null strings, preventing mathematical operations [2, 5].
c) Structural Misalignment: The data isn't in a standard row-column format; some fields (like O105 and C005) appear out of sequence or are clumped together [1, 4].
d) Duplicate Records: Order_ID O101 appears twice.
e) Missing Values: There are explicit Null entries and missing Customer_ID or Sales_Amount links .

Q2. **Primary Key Validation**

Assume Order_ID is the Primary Key.

a) **Is the dataset violating the Primary Key rule?** Yes. The dataset violates the Primary Key rule .

b) **Which record(s) cause this violation?** O101: This Order_ID appears twice in the dataset

Q3. **Missing Value Analysis Which column(s) contain missing values?**

a) List the affected records

- Sales_Amount (Record O102): The data sequence shows O102, C002, then Null where the amount should be.
- Customer_ID (Record O103): The sequence shows O103, then 3200 (Sales_Amount), skipping the Customer_ID field entirely.

b) Explain why loading these records without handling missing values is risky

i-Calculation Errors: Loading Null or missing values into a numeric Sales_Amount field will cause mathematical to fail or produce inaccurate financial reports.

ii-Referential Integrity Breaks: A missing Customer_ID creates an "orphan" record. You cannot link that sale to a specific person, making CRM tracking and customer behavior analysis impossible.

iii-System Crashes: Many database schemas are configured with NOT NULL constraints; attempting to load these records will cause the entire bulk-load process to abort/fail.

iv-Data Ambiguity: Without handling, you cannot distinguish between a zero-dollar sale and a missing data point, leading to poor business decision-making.

Q4. Data Type Validation

Identify records where Sales_Amount violates expected data type rules.

a) **Which record(s) will fail numeric validation?** O102: Contains a Null string value,O105: Contains the text string "Three Thousand"
b) **What would happen if this dataset is loaded into a SQL table with Sales_Amount as Decimal?**
   i-Load Failure/Abort: Most SQL bulk-load operations (like BULK INSERT or COPY) will trigger a Data Type Mismatch error and terminate the process
   ii- Storage Errors: SQL cannot interpret "Three Thousand" as 3000 automatically; it treats it as a VARCHAR, which is physically incompatible with a DECIMAL column structure.

Q5. Date Format Consistency

The Order_Date column has multiple formats

a) **List all date formats present in the dataset .** There are 3 formats :
   i-DD-MM-YYYY: Seen in 12-01-2024 and 15-01-2024.
   ii-YYYY/MM/DD: Seen in 2024/01/18.

iii-Ambiguous DD-MM-YYYY: 20-01-2024 and 25-01-2024 (These follow the first format but confirm the day is leading since there is no 20th or 25th month).

b) **Why is this a problem during data loading?**

i) Parsing Errors: Most database loaders (like SQL Server Integration Services) expect a consistent schema. Encountering a” / “ when expecting “ a” - usually causes the load to crash.

ii) Incorrect Sorting: If the dates are loaded as strings because the system can't reconcile the formats, sorting by date will fail

iii) Logic Flaws: Without a unified format, your time-series analytics will be fundamentally flawed.

Q6. **Load Readiness Decision**

Based on the dataset condition:

a) Should this dataset be loaded directly into the database? (Yes/No) NO

b) Justify your answer with at least three reasons

Reasons are:

i) <u>Integrity Constraint Violations:</u> The duplicate Order_ID (O101) violates the Primary Key rule.

<u>ii)Data Type Incompatibility</u>: The Sales_Amount column contains non-numeric strings like "Three Thousand".

<u>iii) Schema Misalignment:</u> The raw data sequence is corrupted (e.g., O103 is missing a Customer_ID entirely). Loading this without a Transformation (ETL) step will result in "shifted" data.

Q7. Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading.

<u>1-Uniqueness & Nullability (Primary Key Check)</u>

Unique Constraint: Scan Order_ID to ensure no value appears more than once (e.g., catching the O101 duplicate).

Non-Null Check: Verify that no Order_ID fields are empty or contain strings like "Null".

2. DataType & Format Validation

Numeric Verification: Check Sales_Amount for non-digit characters to catch entries like "Three Thousand".

3. Structural & Schema Integrity

Field Count Consistency: Ensure every record has exactly four attributes. This catches cases like O103 where a column was skipped.

4. Range & Logic Sense

Boundary Analysis: Ensure Sales_Amount is not negative and that Order_Date does not exist in the future.

Q8. Cleaning Strategy

Describe the step-by-step cleaning actions required to make this dataset load-ready.

Step 1: Structural Parsing & Normalisation

Action: Flatten the raw string into a consistent table structure (4 columns).

Fix: Manually or programmatically realign shifted records

Step 2: Primary Key De-duplication

Action: Resolve the O101 conflict.

Fix: Use a "Drop Duplicates" function.

Step 3: Numeric Type Casting

Action: Standardise the Sales_Amount column to a Decimal type.

Fix:Convert text-based numbers (e.g., "Three Thousand") to numeric (3000) using a library like Word2Number.

Replace "Null" strings with a 0.00 default or the column mean/median.

Step 4: Date Uniformity

Action: Convert all entries in Order_Date to YYYY-MM-DD.

Step 5: Final Schema Validation

Action: Run a final check against the database schema requirements.

Fix: Ensure all Customer_ID values start with 'C' and all Order_ID values start with 'O'.

Q9. Loading Strategy Selection

Assume this dataset represents daily sales data.

a) **Should a Full Load or Incremental Load be used?** Incremental

b) **Justify your choice.** Because of its efficiency and performance, Scalability

Q10. BI Impact Scenario

Assume this dataset was loaded without cleaning and connected to a BI dashboard.

a) **What incorrect results might appear in Total Sales KPI?**

i-Under-reported Revenue: The total sales figure will be significantly lower than the truth because the BI tool will likely ignore the record containing "Three Thousand" and the Null value, as it cannot include text in a numeric sum.

ii-Double Counting: the duplicate O101 entry will cause that specific sale of 4500 to be counted twice, artificially boosting the total for that day.

iii-Inaccurate Daily Trends: Because of the mixed date formats, the dashboard might fail to group sales correctly by day or month, leading to "gaps" in trend lines or sales appearing in the wrong period.

b) **Which records specifically would cause misleading insights?**

- **O101 (Duplicate): Causes a +4500 error in total revenue calculation due to duplication.**
- **O105 ("Three Thousand"): Causes a -3000 error because BI tools treat non-numeric strings as zero or "null" in a sum.**

- **O102 (Null Sales_Amount): Leads to a missing data point**
- **O103 (Missing Customer_ID): Results in "Unknown" or "Blank" categories.**

c) Why would BI tools not detect these issues automatically?

- **"**Garbage In, Garbage Out" (GIGO): BI tools are designed for visualization, not validation..
- Silent Failures: Most BI software is programmed to be "user-friendly" by skipping errors.
- Lack of Business Context: A BI tool doesn't know that O101 shouldn't exist twice; it simply sees two valid rows of data.